

Question #1: First Question (15 marks): Fill-in the table with the correct answer:

| Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 | Q8 | Q9 | Q10 | Q11 | Q12 |
|----|----|----|----|----|----|----|----|----|-----|-----|-----|
| C  | B  | B  | B  | B  | D  | A  | D  | C  | A   | A   | B   |

1. If the size of memory is 512 bytes, what will be the starting and ending offset addresses of it?

- A. 00H to FFH      B. 000H to 1FFH      C. 0000H to FFFFH      D. 000H to FFFH

2. What is the type of addressing mode used in instruction MOV AL, [BP+6]

- A. Indirect      B. Based relative      C. Direct Indexed      D. Direct

3. the word length of Intel 8086 microprocessor is:

- A. 24      B. 16      C. 20      D. 32

4. 5. If the offset address of X in data segment is equal to 118H, and DS=2000H, then the physical address of X =?

- A. 21118 H      B. 20118H      C. 21117 H      D. 31117H

$20000 + 118$   
 $20118$

5. Which is part of execution unit in Intel 8086 microprocessor?

- A. Segments registers      B. ALU      C. physical address generating unit      D. IP register

6. Show the binary value in register AL after execution the following instruction sequence,

MOV AL, 11H → Ah = 11      0B  
MOV BL, 15H → Bh = 15      0F  
SUB AL, BL → Ah = 0B - 0F →

- A. AL=11111111      B. AL=11111011      C. AL=11111110      D. AL=11111100

7. If the SP register contains the hexadecimal value F11FH, what will be its value after executing the following program segment?

PUSH AX  
PUSH BX  
ADD AX, BX  
POP AX  
POP AX  
PUSH CX

- A. F11FH      B. F11AH      C. F11CH      D. F11DH

8. Find the value of AX after execution the following assembler codes

MOV CX, 5 → CX = 0005  
MOV AX, 0 → AX = 0000

L: ADD AX, CX

DEC CX

CMP CX, 2

JAL L

HLT

A. 9

B. 15

C. 8

D. 12

→  $ADD = 00 + 05 = 0005$        $AX = 0005 + 0004 = 0009$        $AX = 0002$   
→  $CX = 0005 - 1 = 0004$        $CX = 0004 - 0001 = 0003$        $CX = 0002$   
stop

9. Which is not part of execution unit in Intel 8086 microprocessor?

- A. Segments registers      B. ALU      C. physical address generating unit      D. IP register

10. The content of register BL after execution the instruction MOV BL, -11

A. BL=F6H

B. BL=F7H

C. BL=0BHH

D. BL=06H

11. Find the content of AL and zero flag (ZF) and carry flag (SF) after execution of the following code:

MOV AL, 0FDH  
ADD AL, 3  
ADC AL, 0FEH

$\rightarrow Ah = 00$  (CF=1)  
 $\rightarrow Ah = 00 + 1 \rightarrow 0$  (ZF=1)  $\rightarrow$  (SF=1)

A. AL=FFH, ZF=0; SF=1

B. AL=00H, ZF=0; SF=1

C. AL=FFH, ZF=0; SF=0

D. AL=02H, ZF=0; SF=1

12. The offset of a particular segment varies from \_\_\_\_\_ to \_\_\_\_\_

A. 000h to FFFh

B. 0000h to FFFFh

C. 00h to FFh

D. 00000h to FFFFFh

### Question #2 (8 points)

Trace out the following program segments and find registers contents:

|                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>X      X      X      X      X      X      X      X</p> <p>ARR DW 3333H, 4444H, 1111H, 2222H, OFFFH</p> <p>MOV BX, OFFSET ARR      (BX=00)</p> <p>MOV SI, 4      SI=4</p> <p>MOV DI, 2      DI=2</p> <p>MOV DL, [BX][SI]      DL = X+4      (DL=11h)</p> <p>MOV AL, [BX]      <math>\rightarrow Ah = X</math>      (Ah=33)</p> <p>XCHG AL, DL      <math>\rightarrow</math></p> <p>AL = 11h ; DL = 33h</p>        | <p>MOV AL, 22H      AL=22h</p> <p>SUB AL, 22      Ah = 22 - 22 = 0      (ZF=Zero)</p> <p>JZ YES      <math>\rightarrow ZF=1</math></p> <p>SUB AL, 5</p> <p>JMP EE</p> <p>YES: ADD AL, 5      Ah = 0 + 5 = 05</p> <p>EE: HLT</p> <p>AL = 05h</p>                                                                                                                                                                                                                                        |
| <p>ARRAY DB 8, 3, 2, 4, 6, 8, 5, 7, 0</p> <p>MOV AL, 0      Ah=0</p> <p>MOV CX, 4      CX=4</p> <p>MOV BX, OFFSET ARRAY      (BX=00)</p> <p>L:      (BX=00)</p> <p>ADD AL, [BX]      <math>\rightarrow Ah = 0 + 08 = 08</math></p> <p>ADD BX, 2      <math>\rightarrow Bx = X + 2 = 02</math></p> <p>LOOP L      CX=3, CX=2, CX=1, CX=0      (CX=0)</p> <p>MOV AH, [BX]      (AH=15)</p> <p>AL = 15h ; AH = 00h</p> | <p>X DW 104FH, 201AH, C030H</p> <p>MOV BX, OFFSET X      (BX=X)</p> <p>MOV AX, [BX]      <math>\rightarrow AX = 104F</math></p> <p>INC BX      <math>\rightarrow X+1</math></p> <p>MOV CX, [BX]      <math>\rightarrow CX = X+1 = 1050</math></p> <p>PUSH AX      <math>\rightarrow 201A</math></p> <p>PUSH CX      <math>\rightarrow C030</math></p> <p>POP AX      <math>\rightarrow C030</math></p> <p>POP CX      <math>\rightarrow 201A</math></p> <p>AX = C030h ; CX = 201Ah</p> |

### Question #3 (10 points)

a) Suppose we have a microprocessor with 32bits address bus; what is the maximum memory capacity will be?

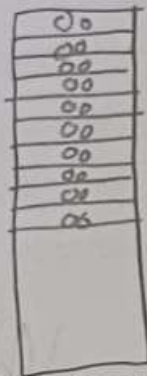
$$= 2^{32} = 2^{30} \cdot 2^2 = 4GB$$

b) For the following instruction MOV [BX], 50H; what is the addressing mode used?

Indirect addressing

- c) Write an assembler codes to store zeroes into X memory locations (3 points)

X DB 10 DUP (?)



- d) Write the equivalent code of the following instructions (2 points)

LOOP L

= ADD CX  
~~INC CX~~  
JNZ = 1

- e) Write an assembler code to implement the following if statement: (3 points)

If  $X < 8$   $s = s + 15$  else  $s = s + 7$ ; assuming x and y are signed bytes.

MOV 100

HLT